



PRODUCT REQUIREMENTS DOCUMENT

ACTIV Digital Membership Platform

Developer Edition - Web, Administration, Android Applications and Platform Services

Product owner	ACTIV	Document version	1.0
Prepared by	Localhost IT Services & Solutions	Status	Development baseline
Prepared on	14 August 2026	Primary audience	Product, design, engineering and QA
Current website	https://activ.org.in/	Platforms	Responsive web and Android

PRODUCT INTENT Deliver one auditable membership journey from discovery and registration through tiered review, plan selection, online payment, activation, events, member discovery and business enablement.

This PRD defines the implementation baseline for the approved functional scope. Values marked TBC require product-owner confirmation before the affected feature is built.

1. Document Purpose and Scope

This document translates the approved ACTIV product scope into testable product and engineering requirements. It is intended to guide UX design, solution architecture, backend and frontend implementation, Android development, integration work and quality assurance.

REQUIREMENT LANGUAGE Must = release-blocking requirement. Should = expected unless a documented trade-off is approved. TBC = decision needed before implementation can be finalized.

1.1 Product Context

ACTIV currently uses its public website to present organisational information, events, membership information, news, forms and contact pathways. The new platform extends that presence into a connected membership system serving applicants, active members and a three-tier administrative hierarchy.

1.2 Goals

- Provide consistent member discovery, registration, review, payment and activation journeys across web and Android.
- Support distinct Company/Business and Aspirant/Student application paths without duplicating the shared account and approval model.
- Give Local/Block, District and State administrators scoped queues and auditable review actions.
- Unlock plan-specific events, directory, catalogue and business capabilities only after verified activation.
- Reduce manual follow-up with event-driven email and WhatsApp notifications.
- Operate through one API-based backend with consistent status, permissions and data rules across all clients.

1.3 Success Criteria

- A new user can complete the correct registration path and submit an application without administrator assistance.
- Every submitted application can be traced through Local/Block, District and State review with actor, timestamp, decision and remarks.
- Only State-approved applicants can initiate membership payment; only a verified gateway result can activate membership.
- Restricted and active features are enforced server-side as well as hidden or disabled in the client interface.

- Critical status changes generate the configured notification and record delivery outcome.
- The same account, application and membership state is presented consistently on member web and Android surfaces.

1.4 Non-Goals

- iOS application development or Apple App Store release.
- Bulk migration of existing member or customer records.
- Advanced BI, predictive analytics, recommendation engines or AI/ML features.
- Unlisted ERP, CRM, accounting, government or third-party system integrations.
- A complete brand-identity exercise or a redesign beyond approved screens and workflows.
- Custom content production, photography, videography, illustration or marketing creative production.
- Long-term infrastructure administration or feature development after handover.

2. Users, Roles and Access Boundaries

Authorization must combine user role, geographic assignment, member type, application status and membership status. Client-side visibility is not sufficient; every protected backend operation must enforce the same rule.

Role	Definition	Primary access
Visitor	Unauthenticated public user	Public pages, onboarding, public events/content, login and registration entry
Applicant	Registered Company/Business or Aspirant/Student user	Own forms, profile completion, application tracking and restricted dashboard
Active member	Activated member with plan entitlements	Member dashboard, events, directory and plan-specific features
Local/Block admin	Administrator assigned to one or more Local/Block units	Applications currently assigned to those units; forward, return or recommend per policy
District admin	Administrator assigned to one or more districts	Applications forwarded from Local/Block within assigned districts
State admin	State-level administrator	Applications forwarded from districts; final approval, rejection or correction decision

2.1 Permission Rules

- A user may read and update only the user's own application and owned business drafts unless granted an administrative role.
- An administrator may act only on applications assigned to the administrator's current level and geographic scope.

- A completed review action is immutable; corrections must be recorded as a new action with a reason.
- State approval is required before payment-order creation.
- Active-member features require an active membership and the matching plan entitlement.
- Administrative roles and geographic assignments must be managed through a controlled process; the role-management owner and UI are TBC.

Capability	Applicant	Active member	Local/Block	District	State
Own registration/profile	Yes	Yes	View	View	View
Submit application	Yes	No	No	No	No
Review assigned application	No	No	Yes	Yes	Yes
Forward to next level	No	No	Yes	Yes	No
Final approval/rejection	No	No	No	No	Yes
Create payment order	Approved user	No	No	No	No
Active member services	No	Entitled	Operational view TBC	Operational view TBC	Operational view TBC

3. Product Surfaces and System Context

Surface	Responsibility	Primary users
Public responsive web	Discovery, ACTIV benefits, public content, member entry points	Visitors and applicants
Member web portal	Registration, profile completion, application tracking, payment and member services	Applicants and active members
Admin web portal	Scoped queues, review, status progression, member lookup and operational audit	Authorized administrators
Member Android app	Mobile version of the agreed applicant and active-member journey	Applicants and active members
Admin Android app	Mobile review and status progression for assigned applications	Authorized administrators
Backend/API	Identity, workflow, membership, payment, events, directory, notifications and audit	All authenticated clients and workers

3.1 Logical Architecture

REQUEST PATH Web clients and Android clients -> authenticated API -> domain services -> relational database and object storage. Domain events -> notification worker -> email/WhatsApp providers. Payment service -> gateway -> verified callback -> membership activation.

- Use one authoritative backend for web and Android; do not implement client-specific business rules that can diverge.
- Represent application review, payment and activation as explicit state transitions with validation guards.
- Use asynchronous workers for notifications and other retryable third-party operations.

- Store uploaded files in private object storage and expose time-limited authorized access rather than public permanent URLs.
- Capture an immutable audit event for privileged actions, security-sensitive changes and payment callbacks.

3.2 Platform Assumptions

- Responsive web and Android are required; exact frameworks and cloud provider are engineering decisions subject to client environment constraints.
- The application will use a relational database because core workflows require transactions, referential integrity and auditable history.
- External credentials and developer accounts will be supplied through a secure channel and must not be committed to source control.
- All timestamps must be stored in UTC and rendered in the configured business timezone.
- Production, staging/UAT and development environments must use separate databases, credentials and provider configurations.

4. End-to-End User Journeys

4.1 Company / Business Journey

1. View onboarding and ACTIV benefits; choose Get Started or sign in.
2. Create an account, verify credentials and select Company/Business as member type.
3. Provide personal and identity details.
4. Provide organisation profile, business categories and operating information.
5. Provide PAN/GST/Udyam/ITR/turnover and government-scheme information as configured.
6. Provide related-concern information, accept the approved undertaking and submit.
7. Use the restricted dashboard to track completeness and review status; prepare business profile and catalogue drafts.
8. After State approval, select the eligible plan and complete online payment.
9. After verified payment, enter the active dashboard and use entitled business, event and discovery features.

4.2 Aspirant / Student Journey

1. View onboarding, create an account and select Aspirant/Student as member type.

2. Provide personal, identity, education and location information.
3. Provide target business domain, mentoring/training needs, goals and the approved undertaking.
4. Submit without Company/Business tax and turnover forms.
5. Track review status through the restricted dashboard.
6. After State approval, select the eligible plan and complete online payment.
7. After verified payment, use the learning, mentoring, networking and eligible event experience.

4.3 Administrative Review Journey

1. System routes a submitted application to the Local/Block queue derived from the applicant's location.
2. Local/Block administrator reviews the complete application and forwards it, returns it for correction or records another permitted decision.
3. District administrator performs the same controlled review for applications forwarded from assigned Local/Block units.
4. State administrator records final approval, rejection or correction request.
5. System exposes the appropriate membership plan only after final approval and notifies the applicant.
6. Every review action records actor, role, geographic scope, prior state, new state, remarks and timestamp.

4.4 Payment and Activation Journey

CRITICAL TRANSITION State approved -> Plan selected -> Payment order created -> Gateway result received -> Server verifies signature/reference -> Payment recorded -> Membership activated -> Entitlements issued.

The browser or Android result screen must never activate membership by itself. Activation occurs only after the backend verifies the gateway callback or performs a trusted server-to-server status check.

5. Functional Requirements

Priorities use Must, Should and TBC as defined in Section 1.

5.1 Discovery and Authentication

ID	Priority	Requirement	Acceptance condition
ONB-001	Must	Present onboarding content explaining networking, business growth, regional associations and industrial catalogues.	User can move through the content and reach Get Started, Login or Skip without a dead end.
ONB-002	Must	Provide responsive public entry points for Login and Registration.	Entry points work on supported mobile and desktop web breakpoints.
AUTH-001	Must	Support email/password account creation and login with show/hide password control.	Invalid email and password rules are explained inline; valid credentials create an authenticated session.
AUTH-002	Must	Prevent duplicate email accounts and normalize email casing.	Registration with an existing normalized email returns a safe, actionable response.
AUTH-003	Must	Provide password reset with expiring, single-use token or link.	A used or expired reset token cannot change the password.
AUTH-004	TBC	Support Google, Facebook and LinkedIn sign-in when approved developer accounts are available.	Enabled providers create or link one platform account without duplicate member records.
AUTH-005	Must	Protect authenticated sessions and permit logout from web and Android.	Expired/revoked sessions are rejected by protected APIs and the client returns to authentication.

5.2 Registration and Profile Completion

ID	Priority	Requirement	Acceptance condition
REG-001	Must	Collect location and member type before displaying pathway-specific forms.	Company/Business and Aspirant/Student users receive the correct form sequence.
REG-002	Must	Save each registration step as a draft and permit resume from the last completed step.	A signed-in user can leave and resume without losing previously saved valid data.
REG-003	Must	Collect Company/Business personal details: full name, identity reference, education, religion, social category and city.	Required-field and format validation runs on client and server; saved values appear on review.
REG-004	Must	Collect organisation name, constitution type, categories, commencement year, employee count and chamber/government memberships.	User can review and amend saved business information before submission.
REG-005	Must	Collect PAN, GST, Udyam registration, ITR status, turnover range and government-scheme benefits when applicable.	Conditional fields appear only under approved rules; server rejects invalid configured formats.
REG-006	Must	Collect sister concerns/company names and record acceptance of the approved legal undertaking.	Submission is blocked until required declarations are accepted and version/time are recorded.
REG-007	Must	Collect Aspirant/Student personal, education, institution, location, target domain, support needs and goals.	Aspirant/Student path omits business tax and turnover forms and reaches submission.
REG-008	Must	Provide a final review screen and explicit submission action.	Submission validates all required data, creates an application snapshot and assigns the initial review state.
REG-009	Should	Show a dynamic completion percentage and links to missing sections.	Percentage is derived from configured required fields and updates after each successful save.

5.3 Restricted Dashboard and Application Tracking

ID	Priority	Requirement	Acceptance condition
PRE-001	Must	Show member name, location, member type, completion status and current application status.	Dashboard reflects the latest authoritative server state after refresh or app relaunch.

ID	Priority	Requirement	Acceptance condition
PRE-002	Must	Explain the current review stage and any applicant action required.	Returned applications show the responsible level, remarks and a clear correction/resubmission action.
PRE-003	Must	Keep premium features locked before activation.	Direct API access to active-only features returns an authorization error for non-active users.
PRE-004	Should	Allow eligible business applicants to prepare logo, description and catalogue drafts.	Drafts remain private and unpublished until entitlement and publication rules are satisfied.
PRE-005	Should	Display a chronological application timeline.	Timeline shows state, timestamp and user-safe description without exposing internal-only remarks.

5.4 Administrative Review

ID	Priority	Requirement	Acceptance condition
ADM-001	Must	Provide role- and geography-scoped application queues with status, member type and date filters.	An admin cannot retrieve an application outside assigned scope, including by direct URL/API.
ADM-002	Must	Display the complete submitted application and attached files required at the current review level.	Admin can inspect the application snapshot without altering applicant-supplied data.
ADM-003	Must	Allow a permitted admin to forward, return for correction or record another approved decision with remarks.	The backend validates current state, role and target transition before saving.
ADM-004	Must	Reserve final approval and rejection for State administration.	Non-State roles receive authorization failure for final-decision endpoints.
ADM-005	Must	Record immutable review history and surface it to authorized admins.	History includes actor, role, scope, prior/new state, remarks and timestamp for each action.
ADM-006	Must	Prevent stale concurrent decisions.	If state changed after the screen loaded, the attempted action is rejected and the latest state is returned.
ADM-007	Should	Provide counts by current queue/status for the signed-in admin's scope.	Counts reconcile with filter results and do not include out-of-scope records.

5.5 Membership Payment and Activation

ID	Priority	Requirement	Acceptance condition
PAY-001	Must	Show only plans eligible for the approved applicant's member type.	Unapproved or mismatched users cannot retrieve or select the plan through the API.
PAY-002	Must	Create a unique server-side payment order containing member, application, plan, amount and provider reference.	Repeated client retries are idempotent and do not create duplicate payable orders.
PAY-003	Must	Open the selected gateway flow for supported methods such as UPI, NetBanking and cards.	Client receives the provider order reference and shows success, pending or failure safely.
PAY-004	Must	Verify callback signature/reference and amount on the server.	Invalid or mismatched callbacks are rejected, logged and cannot activate membership.
PAY-005	Must	Make callback processing idempotent.	Repeated valid callbacks create one payment record and one activation event.
PAY-006	Must	Activate membership and issue entitlements only after verified success.	Membership becomes Active atomically with entitlement issuance or is safely retryable.
PAY-007	Should	Provide payment history and receipt/reference details to the member.	Member can view only the member's own successful and pending transactions.

5.6 Active Member Experience

ID	Priority	Requirement	Acceptance condition
MEM-001	Must	Show an active dashboard with welcome content, association updates, upcoming events and quick actions.	Dashboard content is filtered by active status, member type and configured location rules.
EVT-001	Must	List events with banner, title, type, location, date/time and eligibility.	Member can open an eligible event and see complete details.
EVT-002	Must	Display hourly agenda, speakers, reminders and venue information when supplied.	Event schedule is ordered by start time and rendered in the configured business timezone.
EVT-003	Should	Allow eligible members to register for an event.	Duplicate registrations are prevented and eligibility is checked server-side.
DIR-001	Must	Provide member/company discovery with State, District, Block and sector filters.	Search results respect profile visibility and return only approved public/member-visible fields.
BUS-001	Must	Allow entitled business members to maintain one or more company profiles subject to plan rules.	Unauthorized member types cannot create or update company profiles.
BUS-002	Must	Allow entitled members to create, edit, publish/unpublish and remove product/service catalogue entries.	Published entries require valid ownership and required fields; drafts are not discoverable.
BUS-003	TBC	Provide stock management at the depth approved in detailed design.	Minimum model, adjustment rules, units and audit requirements are confirmed before development.
BUS-004	TBC	Provide basic operational analytics limited to approved screens.	Metrics, source events, date ranges and refresh behavior are documented before development.
ENT-001	Must	Apply plan-specific access for Aspirant/Student and Company/Business members.	Each protected API verifies entitlement; UI explains unavailable features without leaking data.

5.7 Notifications

ID	Priority	Requirement	Acceptance condition
NOT-001	Must	Emit a domain event for registration, submission, return/correction, approval, payment required, payment result and activation.	Each event contains a unique ID and the minimum data required for template rendering.
NOT-002	Must	Send configured email and WhatsApp templates asynchronously.	Primary transaction succeeds independently; delivery attempt is queued and tracked.
NOT-003	Must	Retry transient provider failures with capped backoff and dead-letter handling.	Retry count, last error and terminal status are visible to authorized operations users or logs.
NOT-004	Must	Prevent duplicate notifications for the same event/channel/template version.	Replayed events do not produce duplicate sends beyond documented provider behavior.
NOT-005	Should	Respect configured communication preferences for non-transactional messaging.	Mandatory transactional messages remain distinguishable from optional updates.

5.8 Android Applications and Release

ID	Priority	Requirement	Acceptance condition
AND-001	Must	Provide member Android screens for the agreed onboarding, registration, tracking, payment and active-member flows.	Critical journey behavior matches web and uses the same backend state and permissions.
AND-002	Must	Provide admin Android screens for authentication, assigned queues, application review and permitted decisions.	Admin cannot act outside the same role/scope rules enforced on web.
AND-003	Must	Handle offline/network failure without losing confirmed local form input or showing false success.	Retry behavior is explicit; server state is refreshed after reconnect.

ID	Priority	Requirement	Acceptance condition
AND-004	Must	Produce signed release builds and store-ready metadata/assets supplied or approved by ACTIV.	Release candidate installs, authenticates and completes agreed smoke tests on supported devices.
AND-005	TBC	Set minimum Android OS version and supported device matrix.	Decision is recorded before framework configuration and QA device planning.

6. Business Rules and State Model

State	Meaning	Actor	Permitted next state
Draft	Registration has started but has not been submitted.	Applicant	Submitted
Submitted	Application snapshot created and awaiting routing.	System	Pending Local/Block
Pending Local/Block	Assigned to first administrative tier.	Local/Block admin	Pending District / Returned / Rejected TBC
Pending District	Forwarded from Local/Block.	District admin	Pending State / Returned / Rejected TBC
Pending State	Forwarded from District.	State admin	Approved / Returned / Rejected
Returned	Applicant correction is required.	Applicant	Resubmitted to defined tier
Rejected	Application closed by permitted final decision.	State admin	Appeal/reopen TBC
Approved	Eligible plan may be selected.	Applicant/System	Payment Pending
Payment Pending	Order exists or payment is awaiting completion/verification.	Gateway/System	Active / Payment Failed
Payment Failed	Attempt failed or was cancelled.	Applicant	Payment Pending via retry
Active	Verified payment and entitlements are effective.	System	Suspension/expiry lifecycle TBC

6.1 Transition Guards

- Submission requires all fields marked required for the selected member type and the current form configuration.
- Forwarding requires the application to be in the acting administrator's queue and the action to be valid for that tier.
- Return for correction requires user-visible remarks and a defined destination/resubmission tier.
- Plan selection requires State-approved status and an active plan mapped to the applicant's member type.
- Activation requires verified provider status, matching order amount/reference and an unused successful payment event.
- Every transition uses optimistic concurrency or an equivalent version check to reject stale writes.

6.2 Entitlement Baseline

Member type	Expected access	Restricted unless configured
Aspirant / Student	Individual listing; learning, mentoring and student-oriented events; limited draft ideas as configured	Business tax forms; broad company/catalog privileges; unrestricted product/service publishing
Company / Business	Company profile; product/service catalogue; business discovery; broader event access; approved business tools	Features beyond the selected plan or unapproved operational analytics

6.3 Application Snapshot Rule

Review must operate on a versioned application snapshot. If an applicant edits permitted fields after a return, resubmission creates a new snapshot/version while preserving prior versions and review actions. Active profile edits after activation must not silently rewrite the approved application history.

7. Data Requirements

Entity	Core purpose/data	Classification
UserAccount	Email, credential/provider references, account status, last login	Confidential
MemberProfile	Member type, personal details, location hierarchy, completion data	Confidential / sensitive
BusinessProfile	Organisation details, categories, public description, logo and visibility	Mixed private/public
ComplianceProfile	PAN/GST/Udyam/ITR/turnover and related declarations	Sensitive
Application	Owner, member type, version, current state, assigned geography/tier	Confidential
ApplicationSnapshot	Immutable submitted field values and document references	Confidential / sensitive
ReviewAction	Actor, role, scope, from/to state, remarks and timestamp	Confidential / audit
MembershipPlan	Member type, display metadata, amount, duration and entitlement set	Configured product data
PaymentOrder/Transaction	Order, amount, provider reference, status and verification metadata	Financially sensitive
Membership	Member, plan, status, activation and lifecycle dates	Confidential
Event/Registration	Event details, schedule, eligibility and member registration	Mixed private/public
CatalogueItem/Stock	Owner, product/service data, publication state and optional stock data	Mixed private/public
NotificationLog	Event, channel, recipient reference, template version and delivery status	Confidential
AuditLog	Security-sensitive or privileged operation metadata	Restricted

7.1 Data Validation

- Define required/optional status, allowed values, length and format for every field in the signed-off data dictionary.
- Validate on the server even when equivalent client validation exists.
- Normalize email, phone and identifier formats before duplicate checks while retaining display-safe values where needed.
- Use database constraints for uniqueness and referential integrity where the rule is authoritative.
- Store money in the smallest currency unit and never use floating-point arithmetic for payment totals.
- Version legal undertaking text; store the accepted version, actor and timestamp.

7.2 Sensitive Identity and Compliance Data

DECISION REQUIRED Before collecting Aadhaar or equivalent identity data, ACTIV must approve the legal basis, exact data element, masking, access, retention and deletion rules. Engineering should minimize collection, encrypt approved values and avoid exposing full identifiers in logs, notifications or routine admin lists.

- Mask sensitive identifiers by default and reveal them only to explicitly authorized roles for an approved purpose.
- Encrypt highly sensitive fields at rest in addition to storage-level encryption where feasible.
- Never place PAN, GST, identity numbers, payment tokens or secrets in URLs, analytics payloads or application logs.
- Document retention and deletion rules for rejected, abandoned and inactive records before production launch.

8. Integrations and API Requirements

Integration	Use	Dependency	Engineering concern
Payment gateway	Create order, launch checkout, verify callback/status	Final provider and credentials; Razorpay is indicated in source scope	Signed callbacks, idempotency, reconciliation
Email provider	Transactional templates and delivery status	Sender identity, templates, credentials	Retry, bounce/failure visibility
WhatsApp Business	Transactional status messages	Approved business account, templates, opt-in policy	Template approval and provider throttling
Google/Facebook/LinkedIn	Optional social sign-in	Approved developer apps and redirect URLs	Account linking and provider-policy changes
Object storage	Private documents, logos and catalogue media	Bucket/container and access policy	Private-by-default access, malware/content policy TBC
Play Store	Android release submission	Client developer account, listing content and policy declarations	Review outcome is external
Maps/deep links	Venue directions	TBC: simple external map link or embedded provider	Provider terms and key usage if embedded

8.1 API Conventions

- Use versioned JSON APIs over HTTPS with consistent error envelopes, request IDs and machine-readable error codes.
- Authorize every protected operation using server-side role, ownership, geography, state and entitlement checks.
- Use pagination and bounded filters for lists; never expose unbounded member/application exports through routine endpoints.
- Use idempotency keys for payment-order creation and other externally triggered commands that may be retried.

- Return UTC timestamps in ISO 8601 format and explicit currency codes for money values.
- Do not return sensitive fields unless the endpoint and caller role explicitly require them.

8.2 Suggested Endpoint Groups

Endpoint group	Responsibility
/auth	register, login, social callback, refresh/revoke session, password reset
/me	profile, member type, completion, preferences and entitlements
/applications	draft save, validate, submit, status and version history
/admin/applications	scoped queues, detail, review actions and audit history
/membership-plans	eligible plan discovery
/payments	create-order, verify/callback, status and member history
/members and /companies	authorized directory search and public/member-visible profiles
/events	event discovery, schedule, eligibility and registration
/catalogue	draft, publish, update and remove owned items
/notifications	internal event handling and authorized delivery-status lookup

The exact resource model, request schemas and endpoint names are architecture deliverables. The source scope explicitly references payment order creation and payment verification operations; equivalent versioned routes must be provided even if naming changes.

9. Notification Matrix

Event	Recipient	Channel	Required content	Rule
Account registered	Applicant	Email; WhatsApp TBC	Account created; continue profile	Once per account
Application submitted	Applicant	Email + WhatsApp	Submission reference and next step	Once per application version
Review stage changed	Applicant	Email + WhatsApp	Current stage; no internal-only remarks	On meaningful stage change
Correction requested	Applicant	Email + WhatsApp	User-safe remarks and correction link	Once per return action
Application approved	Applicant	Email + WhatsApp	Approval and plan-selection action	Once per approval
Payment required	Applicant	Email + WhatsApp	Selected/eligible plan and secure payment action	On approved plan availability
Payment success	Member	Email + WhatsApp	Reference and activation status	Once per verified transaction
Payment failed/pending	Applicant	Email; WhatsApp TBC	Safe status and retry/check action	Rate-limited
Membership activated	Member	Email + WhatsApp	Welcome and active dashboard action	Once per activation
New assigned review	Responsible admin	In-app; email/WhatsApp TBC	Queue and application reference	Per assignment, deduplicated
Event reminder	Registered eligible member	Email/WhatsApp TBC	Event, schedule and venue link	Configured offsets

9.1 Template and Delivery Rules

- Templates must be versioned and use approved ACTIV wording, sender identity and contact information.

- Do not include full identity/compliance details, secrets or sensitive payment data in messages.
- Every send attempt must record provider message ID where available, status, timestamp, retry count and last error.
- Notification failures must not roll back a successful application, review, payment or activation transaction.
- Provide a replay mechanism restricted to authorized operations users or engineering support, with duplicate controls.

10. Non-Functional Requirements

The following values are a proposed engineering baseline and should be confirmed during technical kickoff.

ID	Area	Requirement / target	Verification
NFR-SEC-001	Security	HTTPS only; secure headers; secrets outside source control; least-privilege service/database access.	Automated configuration check and manual security review before production.
NFR-SEC-002	Authorization	All protected APIs enforce ownership/RBAC/geography/state/entitlement server-side.	Negative tests cover direct-object and cross-scope access.
NFR-SEC-003	Credentials	Strong password policy, rate limiting and secure password hashing; admin MFA is strongly recommended and TBC.	Authentication abuse tests and configuration review pass.
NFR-PRV-001	Privacy	Data minimization, masking, controlled exports, retention/deletion rules and no sensitive values in logs.	Privacy checklist completed before production data is accepted.
NFR-PERF-001	API performance	Proposed p95: <= 800 ms reads and <= 1500 ms writes, excluding third-party latency, under agreed load.	Load test report for representative critical endpoints.
NFR-WEB-001	Web performance	Target LCP <= 2.5 s on representative public pages under agreed mobile network/device profile.	Lighthouse or equivalent test on release candidate.
NFR-REL-001	Availability	Proposed production target: 99.5% monthly excluding announced maintenance and external-provider outages.	Monitoring and incident evidence can report availability.
NFR-REL-002	Recovery	Proposed RPO <= 24 hours and RTO <= 8 hours; exact backup policy TBC with hosting selection.	Restore procedure is tested in a non-production environment.
NFR-AUD-001	Auditability	Privileged decisions, permission changes and payment verification events are append-only and timestamped.	QA can trace a sample application from submission to activation.
NFR-ACC-001	Accessibility	Responsive web targets WCAG 2.1 AA for core journeys where practical; labels, focus and contrast are verified.	Keyboard and screen-reader smoke tests pass for core web flows.
NFR-COMP-001	Compatibility	Support current major Chrome, Edge and Firefox plus configured Android OS/device matrix.	Browser/device matrix test is completed before release.
NFR-OBS-001	Observability	Structured logs, request IDs, error tracking, health checks and alerting for critical services/workers.	A failed payment callback or notification job can be traced without sensitive-data exposure.

11. Analytics and Operational Telemetry

Operational telemetry is required for reliability and product funnel measurement. Advanced BI and predictive features are not part of this baseline.

- Track onboarding started/completed, registration step completed, application submitted, review transition, plan viewed/selected, payment initiated/result, activation and key active-feature use.

- Use pseudonymous account/application identifiers in analytics; do not include identity, tax, compliance or payment credentials.
- Define one canonical event name, schema and owner for each tracked action across web and Android.
- Keep operational audit logs separate from behavioral analytics because their access, retention and integrity needs differ.
- Dashboard/reporting beyond basic operational counts is TBC and must not delay core transaction logging.

12. QA and Acceptance

12.1 Release-Blocking Journey Tests

Scenario	Acceptance outcome
Company registration	New Company/Business user completes all configured sections, submits and sees Pending Local/Block.
Aspirant registration	New Aspirant/Student user completes the simplified path without business compliance forms and submits.
Correction loop	Admin returns with remarks; applicant corrects and resubmits; history and versions remain intact.
Three-tier approval	Application advances only through permitted Local/Block, District and State actions.
Scope isolation	Admin cannot list, open or act on out-of-scope geography or tier records.
Payment success	Approved applicant pays; valid callback is verified once; membership and entitlements activate.
Payment attack/failure	Invalid signature, amount mismatch, duplicate callback and client-forged success cannot activate membership.
Entitlement enforcement	Inactive or wrong-plan user cannot call active-only APIs or access protected records.
Cross-client consistency	Member web and Android show the same application, payment and membership state.
Notification resilience	Core transaction succeeds when provider fails; retry and delivery status are recorded.
Deployment smoke test	Production web, APIs, workers, database, storage and Android release candidate pass agreed smoke checks.

12.2 Test Levels

- Unit tests for domain rules, validators, state transitions, entitlement checks and payment verification.
- API integration tests using isolated provider test credentials and deterministic callback fixtures.
- Role/permission tests for ownership, geographic scope and every administrative transition.
- Web and Android end-to-end tests for the critical journeys listed above.
- Responsive, browser and Android device-matrix checks for signed release candidates.
- Security, privacy, backup/restore, monitoring and operational-runbook checks before production launch.

12.3 Definition of Done

- Requirement and acceptance condition are implemented and linked to test evidence.
- Server-side validation, authorization, audit and error handling are complete.
- Responsive and Android behavior is verified on the approved compatibility matrix.
- Logs and analytics contain no prohibited sensitive values.
- Configuration, migrations, environment variables and rollback notes are documented.
- Product owner accepts the behavior in UAT with no unresolved release-blocking defect.

13. Release Slices and Handover

Release slice	Included capabilities	Exit signal
Slice A - Web foundation	Public entry, authentication, registration paths, restricted dashboard, admin queues, three-tier review and notifications	Complete submitted application can reach State decision in UAT
Slice B - Payment and member value	Plan eligibility, gateway flow, verified activation, active dashboard, events, directory and business/catalogue features	Approved applicant becomes correctly entitled active member
Slice C - Android and production	Member/admin Android apps, end-to-end regression, production configuration, signing, deployment, store preparation and handover	Production smoke tests and signed Android release candidate pass

13.1 Environment and Deployment Requirements

- Maintain separate development, staging/UAT and production configurations and data stores.
- Run version-controlled database migrations with forward and rollback/recovery notes.
- Use automated or documented repeatable builds for web, backend workers and Android releases.
- Configure domain, HTTPS, storage, database, provider credentials, monitoring and backups before production smoke testing.
- Do not use production member data in development; use sanitized or synthetic test fixtures.
- Record application versions, API versions and Android build/version codes in the release summary.

13.2 Handover Package

- Deployable source and approved production builds for the implemented modules.
- Environment and configuration checklist without embedding secrets.
- Database schema/migration notes and API specification.
- Administrator orientation and concise operational guide.

- Monitoring, backup/restore, provider setup and incident/escalation notes.
- Known limitations, open decisions and release status summary.

14. Open Decisions Before Build Freeze

ID	Decision	Owner
OD-01	Final location hierarchy and how applicant location maps to Local/Block, District and State queues.	Product owner
OD-02	Exact required/optional fields, validation formats and allowed values for both registration paths.	Product + compliance
OD-03	Whether full Aadhaar/identity data is collected; approved legal basis, masking, retention and access policy.	ACTIV legal/compliance
OD-04	Which roles may reject at Local/Block or District and where corrected applications resume.	Product owner
OD-05	Membership plan names, configured amounts, duration, renewal behavior and exact entitlement matrix.	Product owner
OD-06	Final payment provider, supported methods, settlement/reconciliation process and refund/cancellation behavior.	Product + finance operations
OD-07	Approved email/WhatsApp templates, languages, sender accounts and preference/consent rules.	Product + communications
OD-08	Enabled social-login providers and account-linking policy.	Product + engineering
OD-09	Exact scope of stock management, multi-company limits, draft ideas and basic analytics.	Product owner
OD-10	Directory field visibility, member opt-out, search fields and who may see contact information.	Product + privacy
OD-11	Event registration capacity, cancellation, reminder offsets and eligibility rules.	Product owner
OD-12	Minimum Android version, device matrix and notification/deep-link behavior.	Engineering + product
OD-13	Admin role provisioning, deactivation and emergency access process.	ACTIV operations
OD-14	Hosting provider, availability/recovery targets, retention periods and production support ownership.	ACTIV + engineering

BUILD-FREEZE RULE A TBC item may be explored technically, but implementation that would lock its data model, user flow or provider contract should wait for a recorded decision and updated acceptance criteria.

15. Source Traceability

Approved scope area	PRD coverage
Discovery and onboarding	Sections 4.1-4.2; ONB and AUTH requirements
Registration	Sections 4.1-4.2; REG requirements; application data model
Restricted dashboard	PRE requirements; state model
Three-tier review	Administrative journey; ADM requirements; audit model
Membership payment	Payment journey; PAY requirements; integration and state rules
Active dashboard and events	MEM and EVT requirements
Directory and business suite	DIR, BUS and ENT requirements
Email and WhatsApp automation	NOT requirements and notification matrix
Member and admin Android apps	AND requirements
Deployment and Play Store	Release slices, environment requirements and handover

Source basis: ACTIV-Timeline.xlsx, the approved ACTIV digital membership platform proposal scope and the existing ACTIV website context captured during proposal preparation.